
When Does a Fourier Neural Operator Fail? Spectral Diagnostics for Low-Rank Surrogates of Reaction–Diffusion

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Abstract

Neural surrogates for PDE simulators are usually judged by one-step error, which hides the two failures that matter in practice: error accumulation under autoregressive rollout and collapse on unseen parameter regimes. We study a compact CP-factorised Fourier neural operator (LiteFNO) against a parameter-matched CNN on the Gray–Scott benchmark from The Well, asking not only whether the surrogate fails but why. (i) LiteFNO uses 22% fewer FLOPs than the CNN, but both its one-step edge and its rollout advantage in a single seed reverse under three-seed averaging; rollout error is instead governed by the fate of the high-frequency spectrum, which the CNN over-damps and LiteFNO over-excites. (ii) Causally ablating the lowest-magnitude retained Fourier modes shows they contribute little at one step yet dominate mid-horizon rollout error. (iii) In a preliminary leave-one-regime-out study, failure on a held-out Gray–Scott regime correlates with the share of simulator spatial variance at low wavenumber (Spearman $\rho = 0.94$ over six regimes, single seed), a statistic requiring no trained model. We also report noise and 6-bit quantisation robustness, CPU/GPU latency, and a deferral pilot in which ensemble disagreement captures essentially all of the oracle-achievable gain, while showing that out-of-distribution error is spread across steps rather than concentrated in a discardable minority. These results argue for spectral diagnostics, computed from simulator output alone, as a cheap pre-training predictor of surrogate reliability.

1 Introduction

Machine-learned emulators promise to replace expensive simulators, but a surrogate is only useful if it is reliable on the trajectories and regimes a scientist actually queries. Two failure modes recur: *rollout drift*, where small one-step errors compound into unphysical states, and *regime collapse*, where a surrogate degrades sharply on a neighbouring region of parameter space. Both are well known Brandstetter et al. [2022], Lippe et al. [2023], McCabe et al. [2023], yet benchmarks typically report a one-step error blind to them.

We take a diagnostic, representation-level view: an FNO’s truncated spectral weights *are* its representation of the dynamics, and both failure modes can be traced to what that representation keeps and discards. Using the Gray–Scott reaction–diffusion system from *The Well* Ohana et al. [2024], re-simulated across its six qualitative regimes (Appendix Fig. 4), we compare a compact Fourier neural

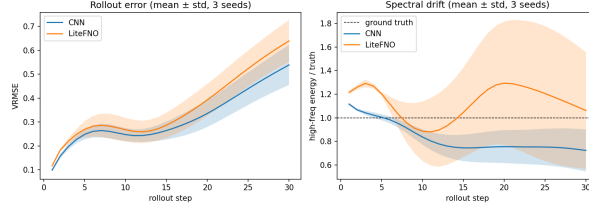


Figure 1: **Headline result (3 seeds).** Left: 30-step rollout VRMSE. Right: ratio of predicted to true energy above wavenumber $k=8$. The CNN steadily loses high-frequency energy; LiteFNO over-excites it with large seed variance. Neither surrogate preserves the spectrum, and the seed-averaged rollout ranking reverses the single-seed result in Appendix Fig. 7.

operator Li et al. [2021] whose spectral weights are CP-factorised (LiteFNO) against a parameter-matched CNN, and ask what in the learned operator, and what in the data, determines when each breaks. Whether a learned representation transfers to a new physical regime, and whether that is predictable *before* training, bears directly on when a trained surrogate can be trusted in place of an imperfect simulator.

Contributions.

1. **An honest efficiency/rollout comparison.** LiteFNO needs 22% fewer FLOPs than the CNN, but both its single-seed one-step and rollout advantages reverse under three-seed averaging (Sec. 3). We report both and show the rollout behaviour is explained by high-frequency spectral drift.
2. **Causal mode ablation.** Zeroing the lowest-magnitude retained Fourier modes of a trained LiteFNO changes one-step error far less than it changes mid-horizon rollout error (which nearly triples), isolating the weak tail of the spectrum as the carrier of rollout drift (Sec. 4).
3. **A pre-training predictor of regime collapse.** In a preliminary leave-one-regime-out study, the held-out/seen error ratio correlates with the fraction of ground-truth spatial variance at low wavenumber ($\rho = +0.94$, six regimes, single seed), a quantity requiring no trained model (Sec. 5).
4. **Deployment and reliability pilots.** Noise and quantisation robustness, batch-1 latency on CPU/GPU as a function of resolution, and a deferral/calibration pilot in which ensemble disagreement matches the oracle but out-of-distribution error cannot be deferred away, and ensembles are overconfident exactly where they are most certain (Sec. 6).

2 Setup

Simulator and data. Gray–Scott models two reacting species (u, v) with feed rate F and kill rate k ; varying (F, k) produces the spots, worms, maze, spirals, bubbles and gliders regimes (Appendix Fig. 4). We use the configuration of *The Well* Ohana et al. [2024], re-simulated with the reference finite-difference solver so that ground-truth energy spectra are available at full resolution. Models are trained on one-step prediction $x_t \mapsto x_{t+1}$ over both fields and evaluated with variance-normalised RMSE (VRMSE, the benchmark’s metric) at one step and over 30-step autoregressive rollouts.

Models. *LiteFNO* is a 2-D FNO with the standard $K \times K$ mode truncation ($K = 16$) whose complex spectral weight tensor is replaced by a rank- r CP decomposition, reducing parameters from $O(C^2 K^2)$ to $O(r(2C + 2K))$. The *CNN* baseline is a residual convolutional network matched to the FNO’s parameter budget. LiteFNO has 8 layers of width 64 with $K=16$ modes and fractional CP rank 0.02; both models are trained on 32^2 fields ($4 \times$ downsampled from *The Well*) with AdamW, learning rate 10^{-3} , StepLR (halved every 100 epochs), batch 64, 200 epochs, MSE loss and seeds 0/1/2 for Secs. 3–4. Our CNN reproduces the published CNN points of *The Well* (Appendix Fig. 6); the published FNO-S sits at $\sim 10^7$ parameters, motivating the compressed variant.

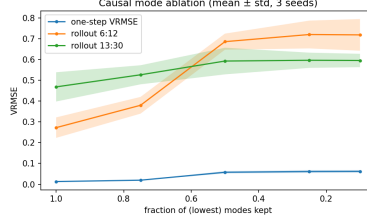


Figure 2: **Causal mode ablation (3 seeds).** Zeroing the lowest-energy fraction of LiteFNO’s Fourier modes at inference leaves one-step error nearly unchanged but sharply increases error in the 6–12 step window, while the late window (13–30) is comparatively insensitive.

3 Efficiency and rollout: a single seed is not enough

Compression. Sweeping CP rank on Gray–Scott (Appendix Fig. 5, left) shows test VRMSE flat from $r=8$ to $r=64$ (≈ 0.03 – 0.04) with a sharp degradation only at $r=128$, where optimisation becomes unstable at the FNO’s native parameter count. Our headline LiteFNO uses fractional CP rank 0.02 (full rank $R=77$; 68 components carry 90% of weight energy across seeds, Appendix Fig. 8a), so the factorisation is well utilised. It costs 1.69×10^8 FLOPs per forward pass against the CNN’s 2.19×10^8 (22% fewer), at $1.8 \times$ the parameter count (180k vs 108k). In a single seed LiteFNO is slightly more accurate at one step (0.0125 vs 0.0152 VRMSE; Appendix Fig. 5, right), but over three seeds the CNN is better (0.0105 vs 0.0132), so we do not claim a one-step accuracy advantage; no Fourier mode is dead (all normalised mode magnitudes exceed 0.7, Appendix Fig. 8b).

Rollout. In a single seed, LiteFNO accumulated error more slowly over 30 steps (0.44 vs 0.62 at step 30; Appendix Fig. 7); three-seed averaging (Fig. 1) reverses this, as at one step: the CNN is better at every horizon (0.54 vs 0.64), with LiteFNO’s seed spread roughly twice as large. We claim no rollout advantage for spectral factorisation; what is stable across seeds is the mechanism in the right panel.

4 Mechanism: truncated modes carry rollout error

Spectral drift. Fig. 1 (right) tracks predicted-to-true energy above $k=8$ along the rollout. The CNN decays monotonically to ≈ 0.7 , low-pass filtering the Turing pattern; LiteFNO overshoots to 1.3 (seed band to 1.8), feeding excess high-frequency energy back and amplifying it. One-step spectra match truth to $k \approx 20$ (Appendix Fig. 9a); the discrepancy is purely a rollout phenomenon.

Causal ablation. To test whether the weakly-weighted modes are causally responsible, we zero the lowest-magnitude fraction of the trained LiteFNO’s modes at inference (Fig. 2). Keeping 10% of modes raises one-step VRMSE from 0.013 to 0.062— $4.7 \times$ relative, yet small in absolute terms next to rollout: over the same ablation, error in steps 6–12 rises 0.28 to 0.72, sharpest between 75% and 50% of modes kept, while the late window (13–30), already dominated by accumulated error, moves only 0.47 to 0.59. The weakest retained modes—far from dead (Appendix Fig. 8b)—carry most of the mid-horizon signal, so mode-count decisions should be made on rollout, not one-step, metrics.

5 Regime generalisation may be predictable from the simulator alone

We train LiteFNO on five Gray–Scott regimes and test on the sixth, for all six folds (Fig. 3). The held-out penalty is wildly uneven: spots and gliders generalise for free (ratio ≤ 1), maze and worms lose little, but spirals and bubbles roughly double in error; the gap does not shrink from epoch 10 to 30 (no budget artefact), and for bubbles is still growing ($1.15 \rightarrow 1.60 \rightarrow 1.87$ —a lower bound).

The right panel explains the ordering: for each regime we compute, from ground truth only, the fraction of spatial variance in Fourier modes with $k < 8$. On the 32^2 grid a 16-mode FNO retains most of the spectrum, so $k=8$ is not the literal truncation boundary but the scale separating coherent structure from pattern-scale detail, chosen before training and untuned. Regimes dominated by large-scale structure (spirals, bubbles) are the ones the surrogate cannot extrapolate to (Spearman



Figure 3: **Leave-one-regime-out.** Left: one-step VRMSE on the five training regimes (green) vs the held-out regime (red). Middle: the held-out/seen ratio does not close with more epochs. Right: the ratio is predicted by the share of ground-truth spatial variance below Fourier mode 8 (Spearman $\rho = +0.94$).

$\rho = +0.94$): the model learns a local rule that transfers among fine-grained regimes, while large-scale coherent structures have no counterpart in the training spectrum. Appendix Fig. 10 gives the per-regime spectra.

Three caveats: this is one seed, four trajectories per regime, and a six-point rank correlation; it *inverts* our pre-registered hypothesis (maze and spots, with the most energy above the cutoff, were predicted hardest and were the two easiest), so the low-mode share is post hoc—a hypothesis generated by these data, not confirmed by them; and VRMSE normalises by per-regime variance, which co-varies with low-mode energy, so the correlation should be re-checked on unnormalised error. If it survives those tests, the predictor is *free*: one short simulation per candidate (F, k) , no training.

6 Deployment robustness and a deferral pilot

Robustness, precision, latency. Both models tolerate input noise to 20 dB SNR (LiteFNO slightly better at 0 dB: 1.15 vs 1.38) and 6-bit weight quantisation, collapsing at 4 bits (Appendix Figs. 12a–12b). Training with matched degradation augmentation halves error at 100% severity but costs 177% on clean inputs and does not transfer to a held-out corruption (Appendix Fig. 13). Parameter count does not predict latency ($\rho=0.07$; the smallest CP model is the slowest at batch 1, 31 ms), but FLOPs do once CP factors are fused into a dense spectral tensor; the fusing speed-up follows the closed-form batch decay but at $1.8\times$ instead of the predicted $12\times$ (Appendix Fig. 14). With grid side N , LiteFNO latency scales as $N^{0.6}-N^{1.1}$ vs the CNN’s $N^{1.3}-N^{1.9}$, favouring the FNO at high resolution; GPU throughput still favours the CNN $6.4\times/\approx 3\times$ at batch 1/16–64 (Appendix Fig. 15).

Deferral and calibration. If regime collapse is detectable at inference, unsafe steps can be handed back to the simulator. In a small pilot (one seed, 4 trajectories/regime, 118 held-out steps, two folds), we rank steps by disagreement of a 4-member ensemble against an oracle deferring the highest-error steps. On bubbles, the fold with a real $1.9\times$ held-out gap, disagreement captures 99–100% of the oracle-achievable gain at every deferral rate (random captures none); the maze fold (Appendix Fig. 11) shows the same tracking but had no gap to close. Less comfortably, on bubbles even 80% deferral leaves error at $1.05\times$ in-distribution—and so does the oracle: OOD error is spread across most steps, so deferral is a detector, not a fix. The same ensemble is overconfident (95% intervals cover 81%/73% in/out of distribution, ECE 0.066/0.131) and worst in its most confident bin: achieved error there is $4.6\text{--}5.2\times$ claimed (Appendix Fig. 16).

7 Related work, limitations, conclusion

Neural operators Li et al. [2021], Kovachki et al. [2023] and factorised variants Tran et al. [2023], Kossaifi et al. [2023] target the trade-off we study; our contribution is the diagnostic. Rollout instability has been attacked via pushforward training Brandstetter et al. [2022], spectral refinement Lippe et al. [2023], and spectral-bias analyses McCabe et al. [2023], Chattopadhyay and Hassanzadeh [2023]; we add a causal ablation tying drift to weakly-weighted retained modes. OOD generalisation of operators is studied in Subramanian et al. [2023], Li et al. [2024]; our low-mode share needs no trained model Ohana et al. [2024]. Deferral is selective prediction Geifman and El-Yaniv [2017].

Limitations and conclusion. All results are on one 2-D system at 32^2 ; only Secs. 3–4 use three seeds, the regime/deferral/calibration studies are single-seed pilots, and the Sec. 5 predictor is a six-

point post-hoc correlation with an open normalisation confound; forced temporal structure (Appendix Fig. 17) will need a temporal analogue of the cutoff. The message: the error that matters hides in the modes one-step evaluation ignores, and a one-line spectral statistic is a candidate predictor of regime coverage.

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A The six Gray–Scott regimes

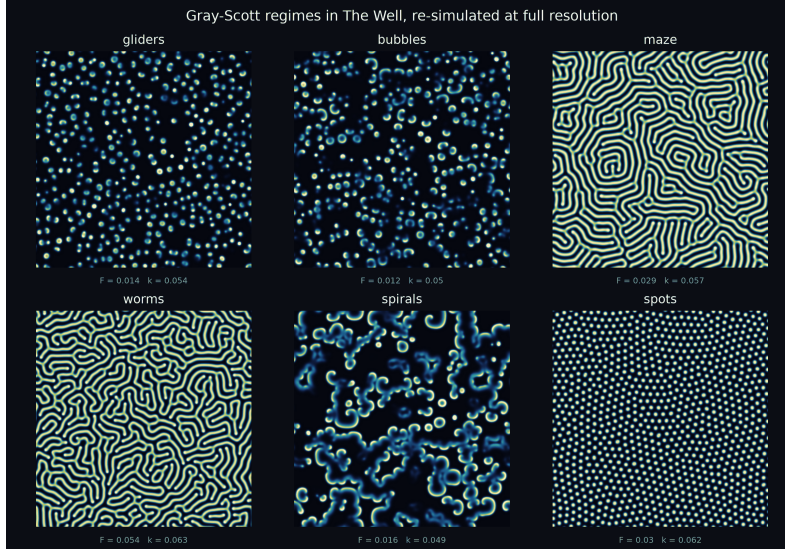


Figure 4: The six Gray–Scott regimes in *The Well*, re-simulated at full resolution with the (F, k) values shown. Regimes differ in their low-wavenumber variance share, a candidate predictor of surrogate generalisation (Sec. 5).

B Reproduction of *The Well* baselines and compression sweep

Figure 6 reproduces the published frontier; Fig. 5 gives the CP-rank sweep and FLOP comparison referenced in Sec. 3; Fig. 7 is the single-seed rollout discussed there.

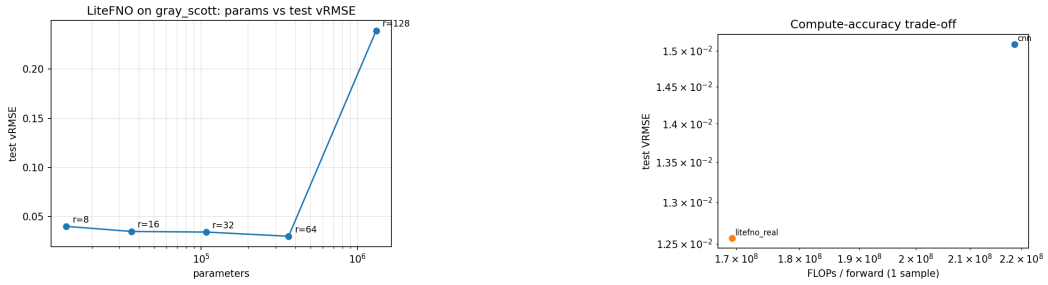


Figure 5: Left: CP-rank sweep; accuracy is flat until $r=128$. Right: single-seed FLOP/accuracy comparison; the one-step ordering reverses over three seeds (Sec. 3).

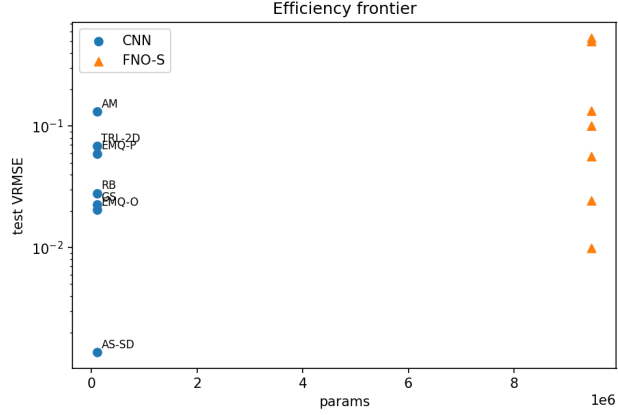


Figure 6: Parameter/accuracy frontier reproduced from *The Well* logs. Our CNN overlaps the published CNN points; the published FNO-S variants lie near 10^7 parameters.

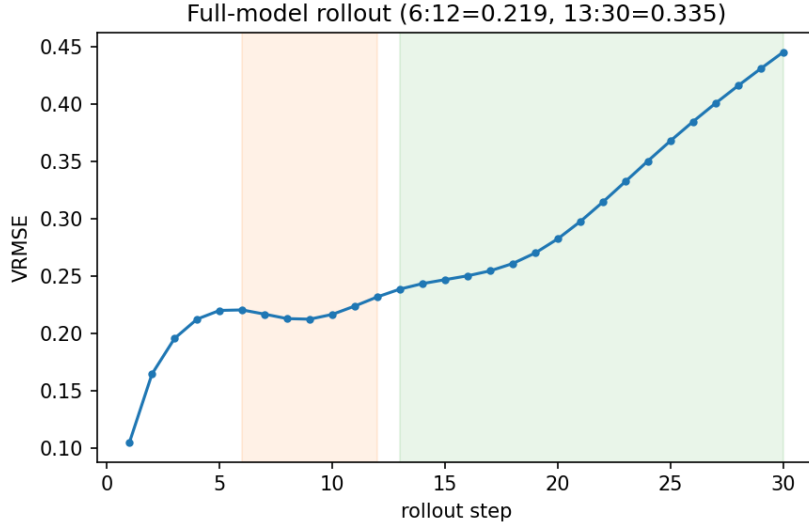


Figure 7: Single-seed rollout and spectral drift; compare with the 3-seed Fig. 1, where the rollout ordering reverses.

C Additional mechanism figures

Figures 8a–10 support Secs. 3–5: the trained LiteFNO uses its full CP rank and all Fourier modes, one-step spectra and error maps are nearly identical across models, and the per-regime spectra give the low-mode share used as the generalisation predictor.

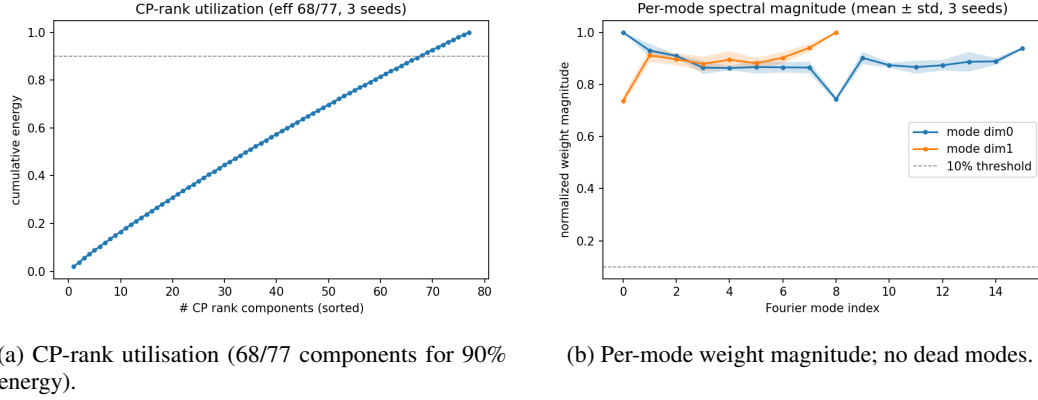


Figure 8: Trained LiteFNO uses its CP rank and all Fourier modes.

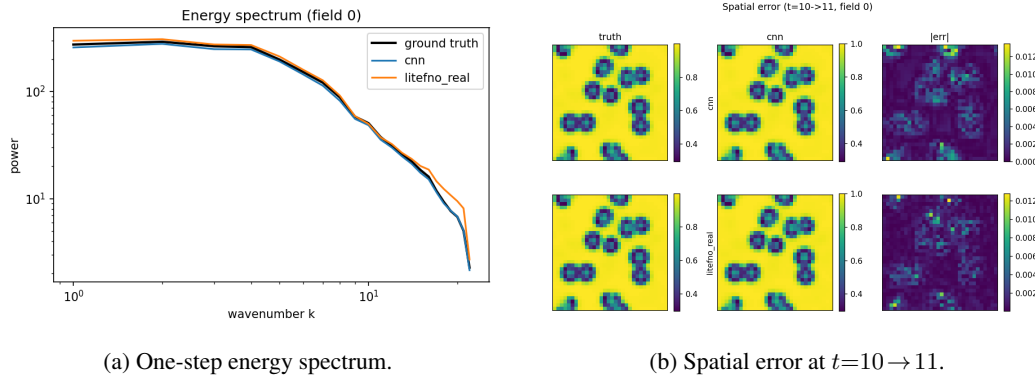


Figure 9: One-step behaviour is nearly identical between models; differences are confined to pattern boundaries.

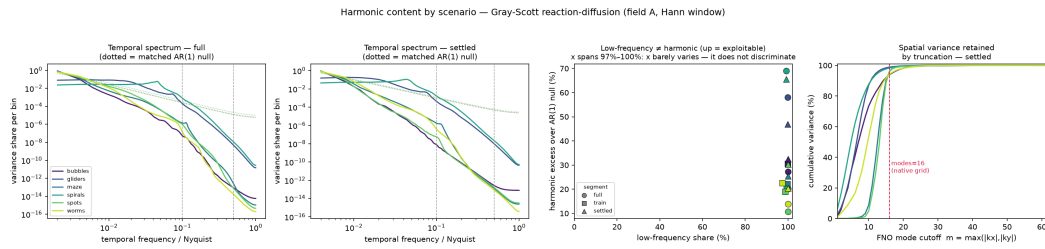


Figure 10: Per-regime spatial and temporal spectra and the variance retained below the mode cutoff used in Fig. 3.

D Robustness and deployment

Figures in this section support Sec. 6.

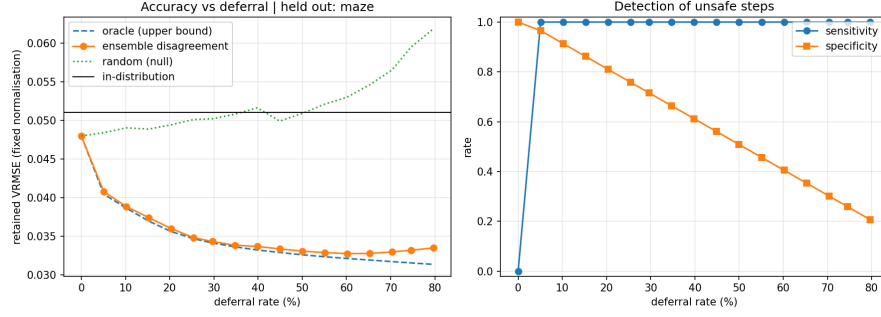
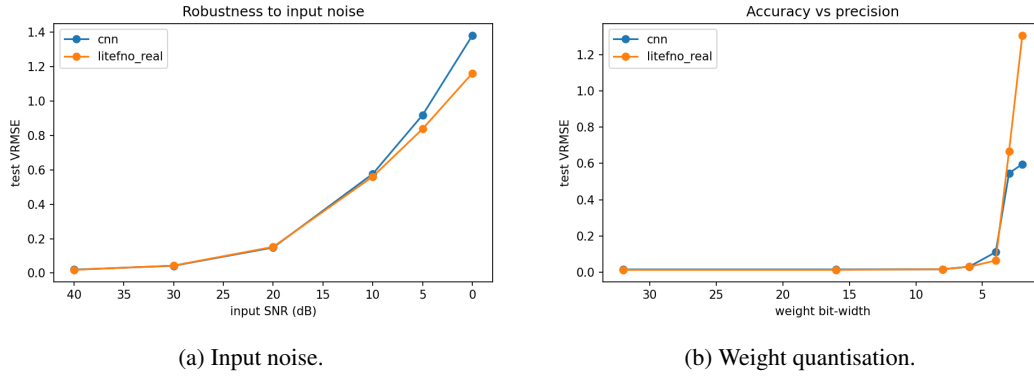


Figure 11: **Deferral pilot (held-out regime: maze)**. Left: retained VRMSE vs fraction of steps deferred; ensemble disagreement tracks the oracle to within 0.002 up to 60% deferral. Note the held-out error already lies below the in-distribution line at 0% deferral on this fold. Right: sensitivity/specificity of flagging unsafe steps.



(a) Input noise.

(b) Weight quantisation.

Figure 12: Both models tolerate 20 dB SNR and 6-bit weights.

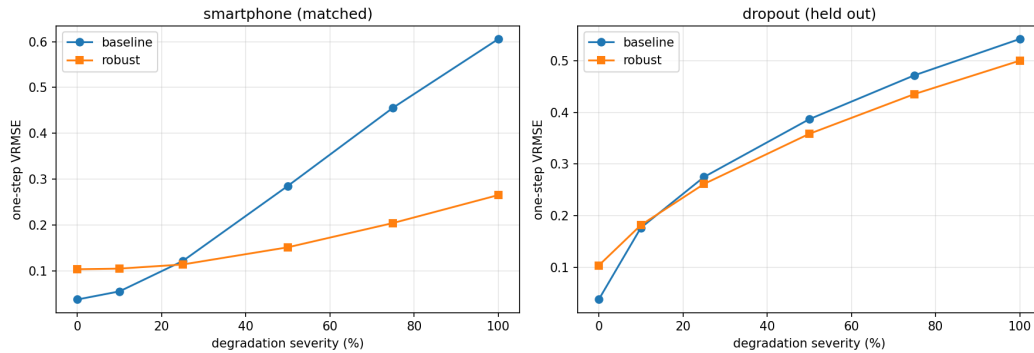


Figure 13: Structured degradation: augmentation-matched training helps on the matched corruption, barely on the held-out one.

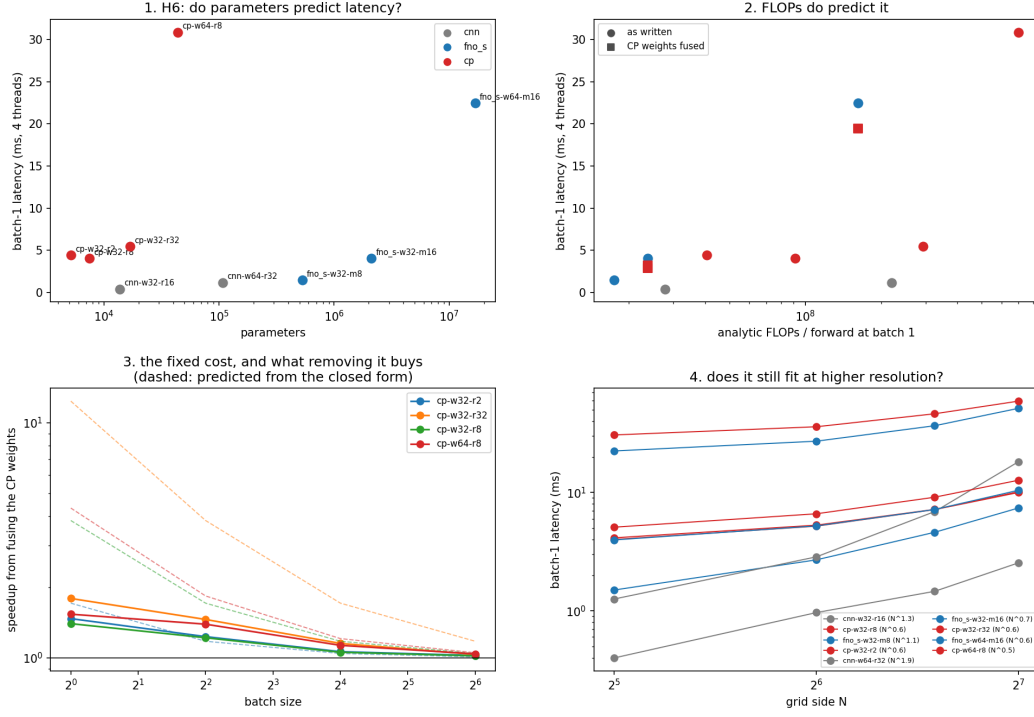


Figure 14: Latency study. (1) Parameters do not predict batch-1 latency; (2) FLOPs do once CP factors are fused; (3) fusing speed-up follows the closed-form batch decay but at a much smaller magnitude; (4) scaling with grid side N .

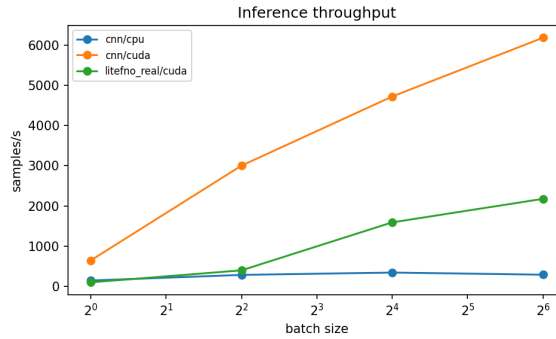


Figure 15: Inference throughput vs batch size.

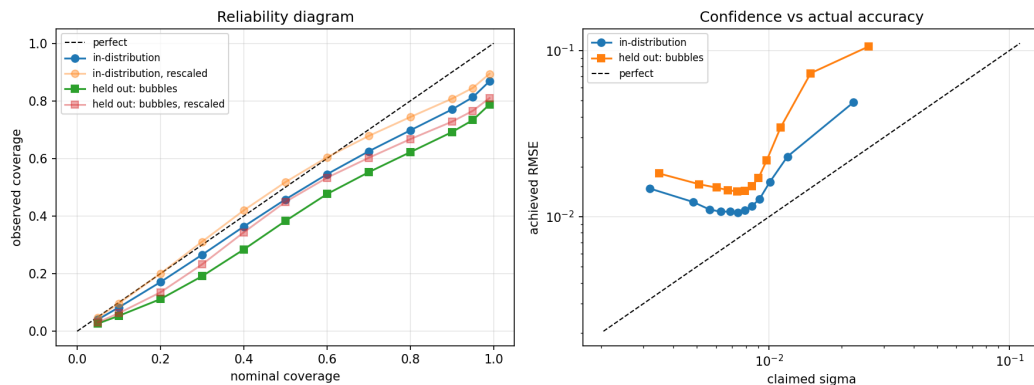


Figure 16: Ensemble calibration, in-distribution and on held-out bubbles: overconfident at all nominal levels, and most overconfident in the highest-confidence bin.

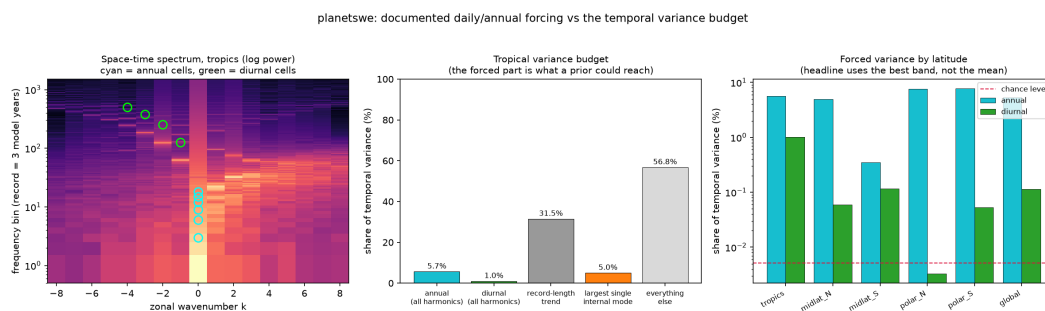


Figure 17: Spatio-temporal variance budget of PlanetSWE; forced diurnal/annual lines dominate, motivating a temporal analogue of the spatial cutoff predictor.